

Digital Control of a 2-DOF AERO System

1 Introduction

The 2-DOF Quanser AERO system consists of two propellers mounted on a base frame that can rotate freely in yaw (circular rotation about the vertical bar) and $\pm 40^\circ$ in pitch (rotation of the horizontal bar). The yaw and pitch motions are the two degrees of freedom (DOF) mentioned in the lab title.



The AERO system has two inputs, which are the voltages to the two motors that drive the propellers. The vertically mounted propeller primarily creates the yaw motion while the horizontally mounted propeller primarily creates the pitch motion. However, the yaw (vertical) propeller also creates some pitching motion and the pitch (horizontal) propeller also creates some yaw. This is an example of what is called a coupled multivariable (or multi-input) system.

2 Regulator Design

The four state variables of the plant are:

$$\begin{aligned} x_1 &= \text{pitch angle (rad)} \\ x_2 &= \text{yaw angle (rad)} \\ x_3 &= \text{pitch rate (rad/sec)} \\ x_4 &= \text{yaw rate (rad/sec)} \end{aligned}$$

The plant inputs are:

$$\begin{aligned} u_1 &= \text{voltage to the pitch motor} \\ u_2 &= \text{voltage to the yaw motor} \end{aligned}$$

The magnitudes of the motor inputs must be less than 24 volts. The state-space model for this plant is given by the following matrices:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \\ -1.7117 & 0 & -0.3249 & 0 \\ 0 & 0 & 0 & -1.0004 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.0377 & 0.0959 \\ -0.1228 & 0.1000 \end{bmatrix}.$$

Notice that $\mathbf{B}$ has two columns, one for each of the plant inputs.

1. Choose a sampling interval and closed-loop pole locations using the guidelines given in class. **The desired settling time is 2 seconds.** Compute the discrete-time (Φ, Γ) ZOH plant equivalent model using `c2d`.
2. Choose s -plane pole locations (`spoles`) and map them into the z -plane (`zpoles`). **Use all of the guidelines given in class for choosing the closed-loop poles of a regulator (i.e. sufficiently damped plant poles, added damping poles, and scaled Bessel poles).** Use `place` to calculate the feedback gain matrix $\mathbf{K}$ for a digital state-feedback regulator. Notice that $\mathbf{K}$ has two rows so that the state feedback $-\mathbf{K}\mathbf{x}$ consists of two signals, one for each motor.
3. Calculate the modern stability robustness bound δ_1 and δ_2 using the function provided on the course web site:

```
[delta1,delta2]=rb_regsf(phi,gamma,K,T)
```

The guideline for these numbers is the following: $\min(\delta_1, \delta_2) \geq 0.5$.

2.1 Simulations

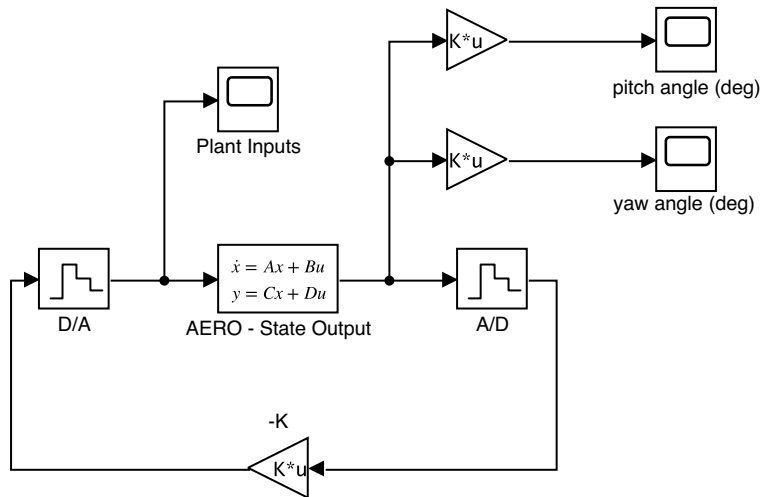


Figure 1: Digital state-feedback regulator for the AERO.

A state feedback regulator will drive all of the state variables to zero. Consider the following situation at time zero: the pitch and yaw angles are zero but someone has tapped the AERO and imparted a nonzero yaw rate. This is modeled with the following initial state vector:

$$\mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.2 \end{bmatrix}.$$

- Simulate the response of the regulator to this initial state and look at the pitch and yaw angles on Simulink scopes. Describe what you would see if you were watching the actual AERO system respond to this initial condition.
- Repeat the simulation with a new initial condition that has only a nonzero pitch rate.

3 Introducing Reference Commands

The simplest way to introduce reference commands into a regulator structure is to create a vector of command signals, $\mathbf{x}_c$, where each variable in $\mathbf{x}_c$ is the command for the corresponding state variable in the plant state vector $\mathbf{x}$. A regulator creates the plant input as follows:

$$\mathbf{u} = -\mathbf{K}\mathbf{x}.$$

If this is changed to the following: $\mathbf{u} = \mathbf{K}(\mathbf{x}_c - \mathbf{x})$, the regulator structure will attempt to drive $\mathbf{x}$ to $\mathbf{x}_c$. Sometimes this works and sometimes it does not work! Section 6.5 in the book discusses this approach to creating a tracking system by introducing reference command into a regulator. **However, the recommended approach to tracking system design is given in Chapter 8, which is the last topic covered in ELE 458.**

Figure 2 shows reference commands for the four state variables. **The pitch command is created by a Signal Generator (in *Sources*) configured to give a square wave with an amplitude of 0.34 and a frequency of 1/15 Hz.** This is put through a digital lowpass filter (LPF) to smooth out the corners of the square wave (use **Discrete transfer function block**; the Matlab code below computes **num** and **den**, the coefficients of the numerator and denominator polynomials). The command for the pitch rate is just the derivative of the command for pitch. Use a **Discrete Derivative block** found in the *Discrete* menu. **The commanded yaw angle is produced by a Sine Wave block (in *Sources*) with Amplitude π , Bias 0, Frequency $2\pi(1/15)$, Phase 0 and Sample Time T.** Given this command for x_2 , what should the command for x_4 be? Because the pitch command has a 20-second period and the yaw command has a 40-second period, you should **change the Simulink simulation time to 40 seconds.**

```
Tsf=2; % filter settling time
den=poly(s2/Tsf);
num=den(end);
LPF=c2d(tf(num,den),T,'tustin');
[num,den]=tfdata(LPf);
num=num{1};
den=den{1};
```

The initial state of the AERO should be set to zero. The r2d gain blocks below convert radians to degrees using a gain of $180/\pi$.

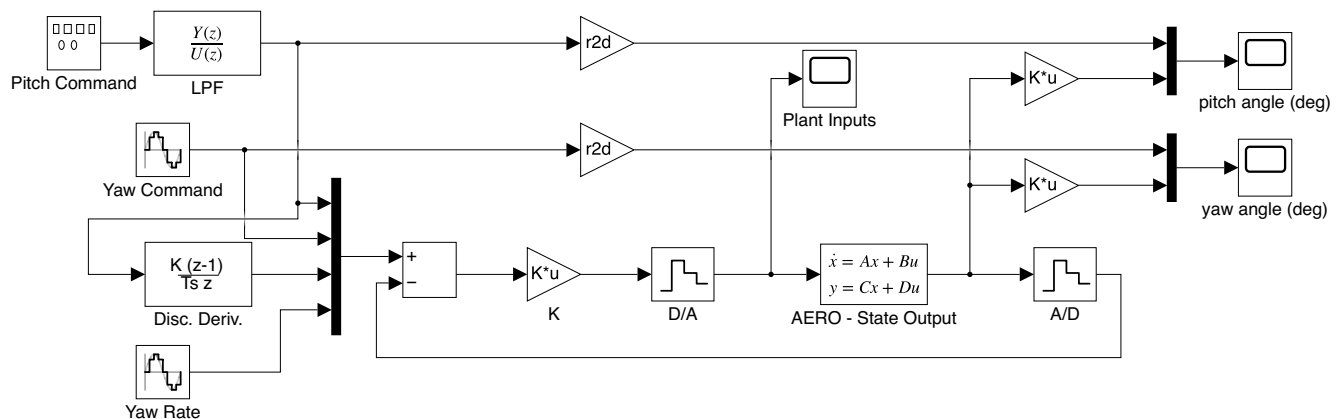


Figure 2: Digital state-feedback regulator with reference commands for the four state variables.

Note that the minus sign on the gain vector $\mathbf{K}$ has been removed from the gain block and moved to the second input of the Adder block. Describe what you would see if you were watching the actual AERO system respond to this control system. **Look at the plant input signals.** How easy (or hard) would it be for a human pilot to create these commands to “fly” this device?